

is also more power efficient. However, it cannot achieve the same blinding intensity of a xenon flash and has its own color cast that is not as pleasing to the eye as the pure white light of a xenon flash.

This is why serious camera phones in the past, particularly from Nokia, have had a xenon flash on them. Because xenon flash stays on for just a fraction of a second, the camera also uses a very, very fast shutter speed. Due to this, you can freeze the action with a xenon flash that you just can't do with an LED flash. Try taking a picture of a moving fan with a camera having a xenon flash and you'll see the blades perfectly stationary in the picture as if the fan was switched off.

Most companies, however, avoid the use of xenon flash on their phones due to the cost and its sheer size compared to the tiny LED. Some compensate for the intensity by having two LED units. This just causes more annoyance to people you try to photograph, as staring into dual LED for full two seconds can leave you dazzled (and not in a good way), unlike a xenon flash that is only on for a split second.

To take care of the color cast, Apple recently introduced dual tone LED flash on the iPhone 5s, which it calls True Tone flash. It uses a white and an amber LED in a pair and depending upon the ambient light the camera smartly adjusts the intensity of each LED unit, thereby trying to get the correct color tone for the flash. This results in a more natural flash color regardless of the situation you are in.

Now that we know enough about the camera software and hardware, we will next learn how to use all that knowledge to capture great photos. 